

Max Rectangle

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1 Description

Given a matrix with unit ('0' or '1') characters values, find the largest rectangle containing 1s.

- **Input:**

```
std::vector<std::vector<char>>> {  
    {'1','0','1','0','0'},  
    {'1','0','1','1','1'},  
    {'1','1','1','1','1'},  
    {'1','0','0','1','0'}}  
};
```

- **Output:**

6

2 Function/Method Declaration

```
class Solution {  
public:  
    int maximalRectangle(vector<vector<char>>& matrix) {  
  
    }  
};
```

3 How to solve

Row 1, Col 1	Row 1, Col 2	Row 1, Col 3	Row 1, Col 4	Row 1, Col 5
□ • h: 1 • d: 1 • w: 1	○ • h: 0 • d: 0	□ • h: 1 • d: 1 • w: 1	○ • h: 0 • d: 0	○ • h: 0 • d: 0
□ • h: 1 • d: 2 • w: 1	○ • h: 0 • d: 0	□ • h: 1 • d: 2 • w: 1	□ • h: 2 • d: 1 • w: 2	□ • h: 3 • d: 1 • w: 3
□ • h: 1 • d: 3 • w: 1	□ • h: 2 • d: 1 • w: 2	□ • h: 3 • d: 3 • w: 1	□ • h: 4 • d: 2 • w: 2	□ • h: 5 • d: 2 • w: 3
□ • h: 1 • d: 4 • w: 1	○ • h: 0 • d: 0	○ • h: 0 • d: 0	□ • h: 1 • d: 3 • w: 2	○ • h: 0 • d: 0

3.1 Algorithm

The above 4 x 5 grid has a small object in the top-left corner of each grid cell(i.e. node) representing a valid (i.e. rectangle) or invalid (circle) cell.

- h - represents horizontal contiguous count. A $h=2$ means there are $h-1$ immediate rectangles to the left including the current node
- d - represents vertical contiguous count. A $d=2$ means there are $d-1$ immediate rectangles to above including the current node
- w - represents maximum horizontal width when calculating max area at current node

3.2 Rules

- $h = 1$; max number of rectangles possible at node : d
- $d = 1$; max number of rectangles possible at node : h
- $d \neq 1$ & $h \neq 1$; max number of rectangles possible at node: $w \times d$